

Python Practice Problems

Exercise 1 — Fundamentals

Topics covered: Strings · Numbers & Math · User Input · Lists & List Functions

20 Problems · 6 Easy · 14 Medium

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Part A —

1. **EASY** **Hello, Name!**

Ask the user to enter their name. Print: Hello, <name>! Welcome to Python.

2. **EASY** **String Length**

Ask the user to type any word. Print the number of characters it contains using `len()`.

3. **EASY** **Repeat a Word**

Ask the user to enter a word and a number `n`. Print the word repeated `n` times on the same line, separated by spaces.

4. **EASY** **Sum of Two Numbers**

Ask the user for two integers. Print their sum, difference, and product — one result per line with a label (e.g. Sum: 15).

5. **EASY** **Even or Odd**

Ask the user for a whole number. Print whether it is **even** or **odd**.

6. **EASY** **Build a List**

Ask the user to enter 5 favourite fruits, one at a time. Store them in a list and print the complete list at the end.

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Part B —

7. **MEDIUM** **Reverse a String**

Ask the user to enter a sentence. Print the sentence reversed (character by character) without using any built-in `reverse()` function — use slicing instead.

8. **MEDIUM** **Count Vowels**

Ask the user to type a sentence. Count and print the total number of vowels (a, e, i, o, u) — both upper and lower case.

9. **MEDIUM** **Title Case Converter**

Ask the user to enter a sentence in all lowercase. Print it in title case (first letter of

every word capitalised) using the appropriate string method. Then also print it in UPPER CASE and lower case.

10. **MEDIUM** **Circle Calculator**

Ask the user for the radius of a circle. Using the `math` library, calculate and print:

- The circumference: $2\pi r$
- The area: πr^2

Round each result to 2 decimal places.

11. **MEDIUM** **Power & Root**

Ask the user for a positive number. Using `math.sqrt()` and the `**` operator, print:

- Its square root
- Its square (number raised to the power of 2)
- Its cube (number raised to the power of 3)

12. **MEDIUM** **Temperature Converter**

Ask the user to enter a temperature in Celsius. Convert and print the equivalent value in both Fahrenheit and Kelvin. Formula: $F = (C \times 9/5) + 32$, $K = C + 273.15$.

13. **MEDIUM** **Shopping List Manager**

Create an empty shopping list. Give the user a simple menu:

- a) Add an item
- b) Remove an item
- c) View the list
- d) Quit

Keep the menu running until the user chooses to quit.

14. **MEDIUM** **List Statistics**

Ask the user to enter 8 numbers, one at a time, and store them in a list. Then print:

- The largest and smallest values using `max()` and `min()`
- The total using `sum()`
- The average (total divided by count)
- The sorted list (ascending)

15. **MEDIUM** **Word Frequency Counter**

Ask the user to type a sentence. Split it into a list of words using `split()`. Count how many times each unique word appears and print each word with its count.

16. MEDIUM Palindrome Checker

Ask the user to enter a word. Check whether it is a palindrome (reads the same forwards and backwards — e.g. `racecar`, `madam`). Print an appropriate message. Ignore case in your comparison.

17. MEDIUM Multiplication Table

Ask the user to enter a number `n` and a range limit `r`. Print the full multiplication table for `n` from 1 up to `r` (e.g. `5 x 3 = 15`), one line per row.

18. MEDIUM List Slicer

Create a list of the integers from 1 to 20. Using list slicing, print:

- The first 5 elements
- The last 5 elements
- Every other element (even-indexed positions)
- The list in reverse order

19. MEDIUM Caesar Cipher

Ask the user for a message and a shift value. Shift every letter in the message forward in the alphabet by the shift amount (wrapping around after `z`) and print the encoded message. Leave spaces and punctuation unchanged. *Hint: use `ord()` and `chr()`.*

20. MEDIUM Number Analyser

Ask the user to enter 10 numbers. Store them in a list and then print:

- How many numbers are positive, negative, and zero
- The absolute value of each number using `abs()` or `math.fabs()`
- The floor and ceiling of each number using `math.floor()` and `math.ceil()`

Tip: Test every program with at least two different inputs — including edge cases like empty strings, zero, or negative numbers.